

⑤ Solve the following LPP by Simplex method max $z = 20x_1 + 6x_2 + 8x_3$

S.T.C $8x_1 + 2x_2 + 3x_3 \leq 250$, $4x_1 + 5x_2 \leq 150$, $2x_1 + x_3 \leq 50$ and

$x_1, x_2, x_3 \geq 0$

Sol

$$z = 20x_1 + 6x_2 + 8x_3 + 0s_1 + 0s_2 + 0s_3$$

$$8x_1 + 2x_2 + 3x_3 + s_1 = 250$$

$$4x_1 + 5x_2 + s_2 = 150$$

$$2x_1 + x_3 + s_3 = 50$$

Iteration - I

	C_j		20	6	8	0	0	0	
	C_B	X_B	x_1	x_2	x_3	s_1	s_2	s_3	min Ratio
1	0	250	8	2	3	1	0	0	$250/8 = 32.25$
2	0	150	4	3	0	0	1	0	$150/4 = 37.5$
3	0	50	<u>2</u>	0	1	0	0	1	$50/2 = 25 \leftarrow$
$z = 0$			0-20	0	0	0	0	0	
			-20	-6	-8	0	0	0	

Iteration - II

$$R_3/2 \Rightarrow \text{New } R_3 \rightarrow R_2 - 4\text{New } R_3, R_1 - 8\text{New } R_3$$

$$\begin{array}{r} 20 \\ \times 25 \\ \hline 100 \end{array}$$



Iteration - II $R_3/2 \Rightarrow \text{New } R_3 \rightarrow R_2 - 4\text{New } R_3, R_1 - 8\text{New } R_3$

	C_j		20	6	8	0	0	0
BV	C_B	X_B	x_1	x_2	x_3	s_1	s_2	s_3
S_1	0	50	0	2	-1	1	1	-4
S_2	0	50	0	(3)	-2	0	1	-2
R_3	20	25	1	0	1/2	0	0	1/2

min Ratio

$$50/2 = 25$$

$$50/3 = 16.6 \text{ min}$$

$$25/0 = \infty$$

$$Z = 500$$

$$\begin{array}{ccccccc} 20 & 0 & 10 & 0 & 0 & 10 \\ -20 & -6 & -8 & -0 & -0 & -0 \end{array}$$

$$\begin{array}{ccccccc} 0 & -6 & 2 & 0 & 0 & 10 \end{array}$$

Iteration - III

	C_j		20	6	8	0	0	0
BV	C_B	X_B	x_1	x_2	x_3	s_1	s_2	s_3
S_1	0	50/3	0	0	1/3	1	1/3	-8/3
R_2	6	50/3	0	1	-4/3	0	1/3	-2/3
R_3	20	25	1	0	1/2	0	0	1/2

$$Z = 600$$

$$\begin{array}{ccccccc} 20 & 6 & 6 & 0 & 2 & 6 \\ -20 & -6 & -8 & -0 & -0 & -0 \\ 0 & 0 & -2 & 0 & 2 & -6 \end{array}$$

$$\begin{array}{r} 20 \\ \times 25 \\ \hline 100 \\ 40 \\ \hline 500 \end{array}$$

$$\begin{array}{r} 50 - 100 \\ 3 \\ \hline 150 - 100 \\ 3 \\ \hline \end{array}$$

$$\begin{array}{r} -1 + \frac{4}{3} \\ -5 + \frac{4}{3} \\ \hline \end{array}$$

$$\begin{array}{r} -4 + \frac{4}{3} \\ -12 + \frac{4}{3} \\ \hline \end{array}$$

$$\begin{array}{r} 1 - \frac{2}{3} \\ 3 - \frac{2}{3} \\ \hline \end{array}$$

$$\begin{array}{r} -4 \\ 10 \\ 6 \end{array}$$

$$\begin{array}{r} -4 \\ -12 \\ 3 \end{array}$$



Iteration - IV

			$R_2 + \frac{2}{3} \text{New } R_1$						
BV	CB	x_B	x_1	x_2	x_3	s_1	s_2	s_3	min/Ratio
x_3	80	50	0	0	1	3	-2	8	
x_2	6	50	0	0	0	2	-1	$\frac{14}{3}$	
x_1	20	0	1	0	0	$-\frac{3}{2}$	3	$-\frac{7}{2}$	
<u>$Z = 400$</u>			20	6	20	72	-66	188	
			-20	-6	-8	-0	-0	-0	

$$\begin{array}{ccccccc} -0 & 0 & 2 & 72 & -66 & 188 \\ \hline \textcircled{1} & & & & & \end{array}$$

$$x_1 = 50$$

$$x_2 = 50$$

$$Z_{\max} = \underline{\underline{700}}$$

Since all $\Delta_j \geq 0$

the current BFS is Optimal

$\therefore$ The Optimal solution is $x_1 = 0$

$$x_2 = 50$$

$$x_3 = 50$$

and $\underline{\underline{Z_{\max} = 700}}$

